



EPE bGP prefix ODN SDN BGP-LS PCE and many more
still missing Service chaining SR-LDP Interworking list
if i miss any

You're absolutely right! There are many more critical Segment Routing concepts that are essential for CCDE-level design. Here's the complete list of all key concepts, including the ones you mentioned plus others that are often missing:

☒ Complete List of Segment Routing Concepts (CCDE-Level)

☒ Core Concepts (Already Covered)

Concept	Already Covered
Segment	✓
SID (Segment Identifier)	✓
SRGB (Segment Routing Global Block)	✓
Node SID / Prefix SID	✓
Adjacency SID	✓
Segment List / Stack	✓
SR-MPLS	✓
SRv6	✓
SR-TE (Traffic Engineering)	✓
TI-LFA (Fast Reroute)	✓
Binding SID	✓
SR Policy	✓

☒ Advanced Concepts (Missing from Earlier)

Q16. What is BGP EPE (Egress Peer Engineering)?

Simple Answer:

BGP EPE = Control which egress peer traffic goes to (not just which egress router).

CCDE-Level Answer:

BGP EPE is Segment Routing for BGP peering — steers traffic to specific BGP peer (not just destination).

Feature	Traditional	With EPE
Destination	Egress router (Node SID)	Egress router + specific peer
Control	BGP best path only	Explicit peer selection
Use case	Reach network	Reach network via specific ISP

How it works:

Peer SID = SID for specific BGP peering (e.g., Router A → ISP X)
Segment List: [Node-A, Peer-SID-ISP-X]

Use Cases:

- **Multi-homing** — send traffic via specific ISP based on cost/SLA
- **Traffic engineering** — optimize egress point selection
- **Cost optimization** — use cheaper transit for certain traffic

BGP-LS Extensions:

EPE SIDs are advertised via **BGP-LS** so controller knows all available peers^{[1] [2]}

CCDE Design Point:

EPE is critical for **service providers** with multiple ISPs — you can optimize egress selection^{[2] [1]}

Q17. What is BGP-LS (BGP Link-State)?

Simple Answer:

BGP-LS = BGP extension that collects network topology (link-state database) and sends it to SDN controller.

CCDE-Level Answer:

BGP-LS is a BGP address family that distributes IGP link-state information (OSPF/IS-IS LSDB) to external entities (SDN controller, PCE).

How it works:

IGP Domain (OSPF/IS-IS) → BGP-LS → Route Reflector → SDN Controller
(collects LSDB) (distributes)

What BGP-LS advertises:

Information	Description
Nodes	Router IDs, loopbacks, capabilities
Links	Interfaces, IP addresses, bandwidth
Prefixes	IP prefixes, Segment Routing SIDs
SR Capabilities	SRGB, SID indices, Flex-Algo
EPE SIDs	BGP peer segments (for Egress Peer Engineering)

Use Cases:

- **SDN Controller** – builds **network-wide topology** ^[3] ^[4]
- **PCE** – computes paths based on full topology ^[3]
- **Multi-domain** – connect multiple IGP domains via BGP-LS ^[3]
- **Traffic Engineering** – controller knows link bandwidth, latency, etc. ^[3]

CCDE Design Point:

BGP-LS is **mandatory** for Carrier SDN – controller needs network-wide visibility to compute optimal paths ^[4] ^[3]

Q18. What is PCE (Path Computation Element)?

Simple Answer:

PCE = Centralized **path computation engine** that calculates optimal paths for SR-TE policies.

CCDE-Level Answer:

PCE is a **network component** (controller) that:

1. Receives **network topology** via BGP-LS
2. Computes **optimal paths** based on constraints (bandwidth, latency, cost)
3. Programs **SR Policies** into headend routers via **PCEP protocol**

How it works:

PCEP (Path Computation Element Protocol)
Headend ↔ PCE (Controller)
(programs SR Policy)

PCE Functions:

Function	Description
Stateful PCE	Maintains state of all computed paths
Stateless PCE	Computes paths on request (no state)
Active PCE	Programs paths into network (SR-TE)
Passive PCE	Only computes, doesn't program

Benefits:

- **Global optimization** — controller sees entire network (not just local view)
- **Constraint-based** — compute paths based on bandwidth, latency, admin policy
- **Dynamic** — recompute paths when network changes

CCDE Design Point:

PCE is centralized control plane — computes paths, headend forwards ^[5] ^[6] ^[3]

Q19. What is ODN (On-Demand Next-hop)?

Simple Answer:

ODN = SDN controller automatically creates SR Policies when it sees new routes.

CCDE-Level Answer:

ODN is an SR Policy automation mechanism where:

- **Headend router signals:** "I need a path to prefix X"
- **Controller** automatically computes path and programs SR Policy
- **No manual configuration** — policy created on-demand

How it works:

Headend: "I need path to 10.0.0.5/32 (ODN template matches)"

Controller: Computes path → Programs SR Policy → Headend steers traffic

Use Cases:

- **Dynamic traffic** — new prefixes don't need manual policy
- **Scalability** — controller creates policies automatically
- **Simplicity** — no manual SR Policy configuration

ODN vs Automated Steering (AS):

Feature	ODN	AS
Trigger	New route seen at headend	Route mapping matches
Template	ODN template (prefix-based)	Mapping (color-endpoint)
Automation	Automatic for new routes	Manual mapping config

CCDE Design Point:

ODN is for **dynamic environments** where prefixes change frequently — controller handles automation^[3]

Q20. What is SDN (Software-Defined Networking) with Segment Routing?

Simple Answer:

SDN + SR = Centralized controller computes paths, network forwards using Segment Routing.

CCDE-Level Answer:

SDN with SR is **centralized control + distributed forwarding**:

Layer	Component	Protocol
Northbound	Controller → Application	REST API, NETCONF
Control	Controller → Router	PCEP, BGP SR-TE
Southbound	Router → Forwarding	SR-MPLS / SRv6
Visibility	Router → Controller	BGP-LS

Controller Architecture:

```
Application Layer (Northbound API)
    ↓
Controller (PCE, BGP SR-TE, ODN)
    ↓
Router (Headend) → SR Policy → Forwarding (SR-MPLS/SRv6)
    ↑
BGP-LS (Topology)
```

Benefits:

- **Centralized optimization** — global view, best paths
- **Automation** — API-driven, no manual config
- **Programmability** — NETCONF/YANG, REST API^[3]
- **Dynamic** — controller reacts to network changes

CCDE Design Point:

SDN + SR is **modern architecture** — controller computes, IGP distributes, network forwards^[3]

Q21. What is Service Chaining with Segment Routing?

Simple Answer:

Service Chaining = Force traffic through **services** (firewall, load balancer, NAT) in specific order.

CCDE-Level Answer:

Service Chaining with SR steers traffic through **service nodes** using segment list:

How it works:

```
Segment List: [Node-A, Service-Firewall, Service-LoadBalancer, Node-B]
               (Go to A)   (Firewall)           (Load Balancer)   (Go to B)
```

Service SID Types:

Service SID	Purpose
Service Node SID	Reach service node (firewall)
Service Index SID	Specific service function
Binding SID	Aggregate multiple services

Use Cases:

- **Security** — firewall → IDS → clean traffic
- **Optimization** — WAN optimizer → compressor
- **Compliance** — traffic must go through inspection

CCDE Design Point:

Service chaining is **SR strength** — no additional protocol needed, just encode service list in segment stack^[1]

Q22. What is SR-LDP Interworking?

Simple Answer:

SR-LDP Interworking = SR and LDP coexist during migration (SR ↔ LDP at boundary).

CCDE-Level Answer:

SR-LDP Interworking allows **gradual migration** from LDP to SR:

How it works:

```
SR Domain ↔ Interworking Gateway ↔ LDP Domain
(SR SIDs)      (maps label ↔ SID)      (LDP labels)
```

Migration Phases:

Phase	Action
1	Enable IGP extensions for SR (OSPF/IS-IS)
2	Enable SR on edge routers first
3	Core runs SR + LDP together

Phase	Action
4	Gradually migrate core to SR-only
5	Remove LDP when all migrated

Benefits:

- **No service disruption** — gradual migration
- **Coexistence** — SR and LDP work together
- **Rollback** — can revert if issues

CCDE Design Point:

SR-LDP interworking is **essential for migration** — plan boundary points carefully^[7] ^[8]

Q23. What is Flex-Algo (Flexible Algorithm)?

Simple Answer:

Flex-Algo = Define custom IGP shortest path (not just metric) — e.g., lowest latency, highest bandwidth.

CCDE-Level Answer:

Flex-Algo allows **multiple IGP topologies** in same network, each with different computation:

How it works:

```
Algorithm 0 (default): Shortest path by IGP metric
Algorithm 128: Shortest path by **delay** (lowest latency)
Algorithm 129: Shortest path by **bandwidth** (highest capacity)
Algorithm 130: Shortest path by **admin group** (exclude certain links)
```

Configuration:

```
Router A:
  Flex-Algo 128: Metric = delay, Exclude-admin = red-links
  Flex-Algo 129: Metric = bandwidth, Include-admin = green-links
```

Use Cases:

- **Low-latency** — real-time traffic (video, VoIP)
- **High-bandwidth** — bulk data transfer
- **Service isolation** — different algorithms for different services
- **Network slicing** — separate topologies per tenant/service

CCDE Design Point:

Flex-Algo is **native network slicing** — no VRFs needed, just different algorithms^[8] ^[2]

Q24. What is BGP SR-TE (BGP Segment Routing Traffic Engineering)?

Simple Answer:

BGP SR-TE = Use BGP (not PCEP) to distribute SR Policies from controller to headend.

CCDE-Level Answer:

BGP SR-TE is alternative to PCEP for SR Policy distribution:

Feature	PCEP	BGP SR-TE
Protocol	PCEP	BGP (new address family)
Distribution	Controller → Headend	BGP Route Reflector → Clients
Scalability	Controller-centric	Route reflection (proven)
Robustness	Single controller	Multiple RR (more robust)
Use case	Smaller networks	Large-scale carrier networks

How it works:

```
Controller → BGP SR-TE NLRI → Route Reflector → Headend Routers
                    (programs SR Policy)
```

CCDE Design Point:

BGP SR-TE is more robust than PCEP — uses proven BGP infrastructure (Route Reflection) ^[3]

Q25. What is Seamless MPLS?

Simple Answer:

Seamless MPLS = Connect multiple IGP domains with BGP (no full-mesh iBGP).

CCDE-Level Answer:

Seamless MPLS is architecture that connects multiple IGP areas using BGP with Prefix SID:

How it works:

```
IGP Area 1 ↔ BGP ↔ IGP Area 2 ↔ BGP ↔ IGP Area 3
(Node SID)  (BGP Prefix SID)  (Node SID)
```

Key Components:

Component	Purpose
BGP Prefix SID	Advertise prefix + SID across domains
BGP-LU	BGP labeled unicast (distributes labels)
ABR	Area Border Router (connects IGP areas)

Benefits:

- No full-mesh iBGP — scales better
- End-to-end LSP — across multiple IGP domains
- SR integrated — BGP Prefix SID for end-to-end path

CCDE Design Point:

Seamless MPLS is carrier architecture — connects multiple IGP domains with BGP + SR ^[9]

Q26. What is SR Authentication (SHA/AH)?

Simple Answer:

SR Authentication = Secure segment routing (prevent unauthorized SIDs).

CCDE-Level Answer:

SR Authentication protects SR control plane:

- IGP authentication — OSPF/IS-IS MD5/SHA for SID distribution
- BGP authentication — MD5/SHA for BGP-LS, BGP SR-TE
- PCEP authentication — secure PCEP session
- SRv6 authentication — IPsec (AH/ESP headers)

CCDE Design Point:

Security is critical for SR — compromised SID can redirect traffic ^[1]

Q27. What is SR Telemetry (sFlow, gNMI, IPFIX)?

Simple Answer:

SR Telemetry = Monitor SR paths (performance, utilization, failures).

CCDE-Level Answer:

SR Telemetry provides real-time visibility into SR paths:

- sFlow/gNMI — collect statistics from routers
- IPFIX — flow monitoring (SR Policy utilization)
- In-band OAM — monitor path performance (latency, loss)
- SR-TE Statistics — per-Policy utilization, packet counts

Use Cases:

- Performance monitoring — latency, jitter, loss
- Capacity planning — link utilization
- Troubleshooting — verify SR Policy forwarding

CCDE Design Point:

Telemetry is essential for SR-TE — controller needs real-time data for optimization ^[3]

☒ Complete List of All Segment Routing Concepts

#	Concept	Already Covered?
1	Segment	✓
2	SID (Segment Identifier)	✓
3	SRGB	✓
4	Node SID / Prefix SID	✓
5	Adjacency SID	✓
6	Segment List / Stack	✓
7	SR-MPLS	✓
8	SRv6	✓
9	SR-TE	✓
10	TI-LFA	✓
11	Binding SID	✓
12	SR Policy	✓
13	BGP EPE	✓ (new)
14	BGP-LS	✓ (new)
15	PCE	✓ (new)
16	ODN	✓ (new)
17	SDN + SR	✓ (new)
18	Service Chaining	✓ (new)
19	SR-LDP Interworking	✓ (new)
20	Flex-Algo	✓ (new)
21	BGP SR-TE	✓ (new)
22	Seamless MPLS	✓ (new)
23	SR Authentication	✓ (new)
24	SR Telemetry	✓ (new)
25	AS (Automated Steering)	✓ (new)
26	uSID (micro-SID)	✓ (partial)
27	SR Policy Color/Endpoint	✓ (new)
28	MSD (Maximum SID Depth)	✓ (partial)
29	PCEP (Path Computation Element Protocol)	✓ (new)
30	SR Multi-Domain	✓ (new)

☒ CCDE Interview Answer Summary

Interviewer: "What advanced Segment Routing concepts are critical for carrier networks?"

Your Answer:

"Beyond core concepts (SID, Node/Adj SID, SR-MPLS/SRv6), critical advanced concepts include:

1. **BGP EPE** — Control egress peer selection (not just router)
2. **BGP-LS** — Distribute topology to SDN controller (network-wide visibility)
3. **PCE** — Centralized path computation (global optimization)
4. **ODN** — Automatic SR Policy creation for new prefixes
5. **SDN + SR** — Centralized control + distributed forwarding
6. **Service Chaining** — Steer traffic through services (firewall, load balancer)
7. **SR-LDP Interworking** — Gradual migration from LDP to SR
8. **Flex-Algo** — Custom IGP algorithms (low-latency, high-bandwidth)
9. **BGP SR-TE** — BGP-based SR Policy distribution (more robust than PCEP)
10. **Seamless MPLS** — Multi-domain architecture with BGP Prefix SID

These enable Carrier SDN, automation, network slicing, and optimized traffic engineering."

[6] [2] [8] [3]

Would you like **detailed configuration examples** for any of these concepts (e.g., "BGP EPE configuration on Cisco IOS XR", "Flex-Algo setup for low-latency paths", or "ODN with SDN controller"), or **CCDE design scenarios** (e.g., "Design Provider network with SR-EPE, BGP-LS, and PCE").

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1. <https://datatracker.ietf.org/doc/draft-ietf-idr-bgpls-segment-routing-epe/17/>
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